

Mathematical background

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Chapter 2 rewrite (plan v1.2)
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Abstract

Background material for the thesis: Markovian language, branching criticality, quasi-stationarity, and analytical tools for continuous-time population models. This draft is being rewritten in stages against a frozen structure plan (Paths A–C).

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1 Introduction and roadmap

Deterministic rate equations describe a population well once its numbers are large, and badly once they are not. When counts are of order one—in early infection, and in small-population regimes generally—extinction is an event of genuine probability rather than a limiting curiosity, and the typical size of those lineages that are still alive can differ sharply from the unconditional mean. Branching processes supply the natural stochastic language for that setting, because they track whole lineages rather than an averaged density. This chapter develops the background used throughout the thesis—Markovian population models, quasi-stationarity, and analytical tools for continuous-time generating functions—and then, at greater length, the discrete-time constant $A(p)$ that organises subcritical Galton–Watson behaviour conditional on survival.

The material is arranged in three reading paths, sharing one chapter numbering; the split concerns only what a given reader must carry forward.

Path A. Shared core (preliminaries through checkpoint). Definitions and results every thesis reader should take: a short Markov/probability generating function toolkit; discrete branching and the definition of $A(p)$; continuous-time birth–death quasi-stationarity, including the continuous analogue of $A(p)$; survivability for small initial cohorts; and a brief checkpoint. Conditional laws and quasi-stationary means are introduced where they first arise rather than stockpiled in the preliminaries.

Path B. Method of characteristics. A self-contained analytical ladder for probability generating functions of continuous-time absorption models (with and without birth and death). It sits after the core because it is a full technical development, not because it is dispensable: readers who will work with those probability generating function partial differential equations should treat Path B as required. Catastrophe structure for full BDC models is developed in later chapters.

Path C. The constant $A(p)$ and the Koenigs obstruction. Optional depth: numerics and representations for $A(p)$, the absence of a known elementary closed form, and the link to the Koenigs function of the logistic map. The core will already *state* that no comparably elementary closed form for discrete $A(p)$ is available, while the constant itself remains well defined and computable; Path C supplies the representations, the numerical evidence, and the mechanism behind that gap.

A short chapter appendix holds the heavy derivations—discrete moments, long characteristic calculations, Koenigs detail—each labelled by the path it supports.

Two classical regimes will recur. In slightly supercritical branching, the chance of eventual survival is highly sensitive to the initial number of individuals, a fact with an obvious reading for early infection. In the subcritical regime extinction is certain, and yet the population size conditional on non-extinction can settle into a quasi-stationary picture; the reciprocal of that conditional mean is explicit for continuous-time birth–death, whereas in discrete time it is encoded by $A(p)$. The discrete factor is the analytically harder of the two, and the chapter returns to it in full only in Path C.

What follows is written so that a reader with standard undergraduate probability and ordinary differential equations can proceed without the rest of the thesis in hand. Forward references to later chapters are kept minimal; the aim is a reusable stochastic structure rather than a catalogue of which later section cites which formula.

2 Mathematical preliminaries

This section fixes a short common vocabulary. The aim is not a course in stochastic processes but a set of names and pictures that the rest of the chapter—and the continuous-time models later in the thesis—can use without redefinition. We assume standard undergraduate probability and ordinary differential equations, and concentrate on three things: discrete- and continuous-time Markov structure, generating functions, and the elementary trichotomy of branching criticality.

2.1 Discrete-time Markov chains

A discrete-time Markov chain on a countable state space $\mathcal{S}$ is a sequence of random variables $(\mathbf{X}_n)_{n \geq 0}$ taking values in $\mathcal{S}$ and satisfying the Markov property: the conditional law of the next state depends on the past only through the present. Writing $p(i, j)$ for the one-step transition probability from i to j , that property reads

$$\Pr \{ \mathbf{X}_{n+1} = j \mid \mathbf{X}_n = i, \mathbf{X}_{n-1} = i_{n-1}, \dots, \mathbf{X}_0 = i_0 \} = \Pr \{ \mathbf{X}_{n+1} = j \mid \mathbf{X}_n = i \} = p(i, j),$$

whenever the conditioning event has positive probability.

A state $a \in \mathcal{S}$ is *absorbing* if $p(a, a) = 1$; once the chain enters such a state it remains there forever. Extinction of a population, and later the “rupture” or catastrophe events of birth–death–catastrophe models, will all be encoded by absorption of this kind, sometimes after enlarging the state space with a cemetery state.

The Galton–Watson process treated later in this chapter is a discrete-time Markov chain of exactly this type: it lives on $\mathcal{S} = \{0, 1, 2, \dots\}$ and is absorbed at 0. Its one-step transitions are determined by an offspring distribution rather than by an arbitrary matrix $p(i, j)$, but the Markov and absorption language carries over unchanged.

2.2 Continuous-time Markov chains

A continuous-time Markov chain (CTMC) on a countable state space replaces discrete steps by holding times and jumps. While the process occupies state i it waits an exponential time whose rate q_i may depend on i , and then jumps to a different state j with probabilities determined by the off-diagonal rates $q(i, j)$. The infinitesimal description, for small $\Delta t > 0$, is

$$\begin{aligned} \Pr \{ \mathbf{X}_{t+\Delta t} = j \mid \mathbf{X}_t = i \} &= q(i, j) \Delta t + o(\Delta t), & j \neq i, \\ \Pr \{ \mathbf{X}_{t+\Delta t} = i \mid \mathbf{X}_t = i \} &= 1 - q_i \Delta t + o(\Delta t), \end{aligned}$$

with $q_i = \sum_{j \neq i} q(i, j)$, assumed finite. The Markov property again holds, now with respect to continuous time.

In population models these rates are typically linear, or at least affine, in the current count. A single-type birth–death process on $\{0, 1, 2, \dots\}$ with per capita birth rate β and death rate μ therefore has, when $\mathbf{X}_t = n \geq 1$,

$$q(n, n+1) = \beta n, \quad q(n, n-1) = \mu n, \quad q_n = (\beta + \mu)n,$$

with 0 absorbing if there is no immigration. A catastrophe or “killing” mechanism adds a further transition out of every $n \geq 1$, at a rate proportional to n or to some more general function of n ; that structure is the continuous-time backbone of the birth–death–catastrophe analysis later in the thesis.

Two elementary facts will be used repeatedly. First, the holding time in i is memoryless, so the future after a jump depends only on the arrival state. Second, absorption is again well defined: if $q_a = 0$, then a is absorbing in continuous time. Pathwise, a CTMC may be constructed from its jump chain—the discrete-time sequence of visited states—together with independent exponential clocks. We shall seldom need that construction explicitly, but it is worth having in view, since it explains why discrete-time branching intuition and continuous-time rate models sit in the same family.

2.3 Probability generating functions

For a random variable N taking values in $\{0, 1, 2, \dots\}$, the *probability generating function* (probability generating function) is

$$G(z) := \mathbb{E}(z^N) = \sum_{k=0}^{\infty} \Pr\{N = k\} z^k, \quad |z| \leq 1.$$

When the series converges in a neighbourhood of the origin, derivatives at 0 recover the point masses, and

$$\mathbb{E}(N) = G'(1), \quad \mathbb{E}(N(N-1)) = G''(1),$$

whenever those moments exist, with one-sided derivatives at $z = 1$ if necessary.

The reason generating functions earn their place here is that they turn independent sums into products, and thereby turn many branching calculations into functional iteration or ordinary differential equations. If each individual in a Galton–Watson generation produces offspring independently with generating function $f(z)$, then the population size after one generation from a single ancestor has generating function f , and after n generations $f^{\circ n}$. In continuous time the probability generating function of a CTMC's state, $G(z, t) = \mathbb{E}(z^{\mathbf{X}_t})$, often satisfies a first-order partial differential equation obtained from the master equation (Section 2.5). That second route is opened for linear birth–death in the continuous-time section below and developed fully by the method of characteristics later in the chapter.

2.4 Criticality for branching

Branching mechanisms are organised by a single comparison of mean growth to replacement. The same trichotomy appears in discrete and continuous time; only the bookkeeping of the mean changes.

Discrete time. Suppose each individual has offspring mean $m = \mathbb{E}(\xi)$, where ξ is a generic offspring count, and exclude the trivial law concentrated at 1. The associated Galton–Watson process is

- *subcritical* if $m < 1$,
- *critical* if $m = 1$,
- *supercritical* if $m > 1$.

In the binary death-or-division case used throughout much of this chapter, an individual is replaced by two offspring with probability p and by zero with probability $1 - p$, so that $m = 2p$. Subcriticality is then $p < \frac{1}{2}$, criticality $p = \frac{1}{2}$, and supercriticality $p > \frac{1}{2}$.

Continuous time. For a linear birth–death process with per capita rates β and μ , the mean $M(t) = \mathbb{E}(\mathbf{X}_t)$ obeys $\frac{dM}{dt} = (\beta - \mu)M$ when it exists, so the net growth rate $\beta - \mu$ plays the role of $\log m$ per unit time. The process is called subcritical, critical, or supercritical according as $\beta - \mu$ is negative, zero, or positive, again excluding degenerate cases with no randomness.

Why the trichotomy governs what follows. Extinction probabilities, the existence of a non-trivial survival event, and the nature of population size *conditional on non-extinction* all change character across the critical threshold. In the subcritical regime extinction is certain, yet the conditional distribution of size given survival can settle into a quasi-stationary picture; near criticality those conditional sizes become large; above criticality there is a positive probability of indefinite growth. (A brief discussion already appears in Chapter 1; the definitions here are self-contained, so that the present chapter can be read on its own.) The sections on discrete and continuous branching specialise these regimes in turn, and the constant $A(p)$ of the discrete theory will live naturally in the subcritical half-line $p \in [0, \frac{1}{2})$.

2.5 Forward equations and the master equation

Let $p_n(t) = \Pr \{\mathbf{X}_t = n\}$ for a CTMC on $\{0, 1, 2, \dots\}$. Balancing probability flux into and out of each state yields the *forward Kolmogorov equations*, often called the *master equation* in the physical literature. For the linear birth–death process with rates β, μ as above they read

$$\begin{aligned} \frac{dp_0}{dt} &= \mu p_1, \\ \frac{dp_n}{dt} &= \beta(n-1)p_{n-1} + \mu(n+1)p_{n+1} - (\beta + \mu)n p_n, \quad n \geq 1, \end{aligned}$$

with the usual convention that terms with negative indices are omitted. The first line records that the only way to enter 0 is by a death from state 1; the second records births from $n-1$, deaths from $n+1$, and the total exit rate from n .

As it stands the master equation is an infinite linear system for the law of $\mathbf{X}_t$, and two standard compressions of it will matter later. Taking moments against n produces ordinary differential equations for means and, with more work, for higher moments. Passing instead to the probability generating function $G(z, t) = \sum_n p_n(t)z^n$ converts the same system into a single first-order partial differential equation for G , which is the natural continuous-time counterpart of the functional iteration of discrete branching. We do not develop that partial differential equation here; it appears first for linear birth–death in the continuous-time section of this chapter, and then in greater generality under the method of characteristics.

3 Discrete-time branching

We now specialise the Markov language of Section 2 to the Galton–Watson process, and introduce the constant $A(p)$ that organises subcritical behaviour conditional on survival. The full analytic treatment of $A(p)$ —product formulae, numerical curves, and the Koenigs obstruction—is deferred to Path C. What is needed here is only the definition, the quasi-stationary mean it controls, and a clear statement of which parts of the picture are elementary and which are not.

3.1 Galton–Watson process

Suppose each individual produces, independently of all others, a random number of offspring ξ with

$$\Pr \{\xi = k\} = p_k, \quad k = 0, 1, 2, \dots,$$

and mean $m = \mathbb{E}(\xi)$; we reserve $\mathbf{X}, \mathbf{Z}$ for process state, not for a single offspring draw. Let $\mathbf{Z}_n$ denote the population size in generation n , starting from $\mathbf{Z}_0 = 1$, so that $\mathbb{E}(\mathbf{Z}_n) = m^n$. If $m < 1$, and excluding the degenerate law concentrated at 1, extinction is certain:

$$\lim_{n \rightarrow \infty} \Pr \{\mathbf{Z}_n = 0\} = 1.$$

Write $\phi(z) = \mathbb{E}(z^\xi)$ for the offspring probability generating function, and $u_n = \Pr \{\mathbf{Z}_n = 0\}$ for the extinction probability by generation n from a single ancestor—written ${}_1u_n$ when the initial count must be stressed. Conditioning on the first generation yields the iteration

$$u_{n+1} = \phi(u_n), \quad u_0 = 0, \tag{1}$$

which is the discrete engine driving everything that follows.

3.2 Death or division

Throughout this section we take the binary law

$$\begin{aligned} p_2 &= p && \text{(division),} \\ p_0 &= 1 - p && \text{(death),} \end{aligned}$$

with $p_k = 0$ otherwise. Then $m = 2p$, so the process is subcritical, critical, or supercritical according as $p < \frac{1}{2}$, $p = \frac{1}{2}$, or $p > \frac{1}{2}$, and the unconditional mean is

$$\mathbb{E}(\mathbf{Z}_n) = (2p)^n.$$

Under this law equation (1) becomes

$$u_{n+1} = 1 - p + p u_n^2. \tag{2}$$

It is more convenient to follow survival than extinction, so let $S_n = 1 - u_n$ be the probability of survival to generation n . From (2) one obtains

$$S_{n+1} = 2p S_n - p S_n^2. \tag{3}$$

3.3 Late-time survival and the constant $A(p)$

Restrict attention to the subcritical regime $p < \frac{1}{2}$, where both the mean $(2p)^n$ and the survival probability S_n tend to zero. Because S_n is small at late times, the quadratic term in (3) becomes negligible relative to the linear one, and

$$S_{n+1} \sim 2p S_n.$$

The decay is therefore geometric at ratio $2p$, and the only question left is the prefactor. More precisely, there exists a constant $A(p) \in (0, \frac{1}{2}]$, depending on p but not on n , such that

$$S_n \sim A(p) (2p)^n \quad \text{as } n \rightarrow \infty, \tag{4}$$

or equivalently

$$A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n}. \quad (5)$$

(Existence and positivity of the limit are classical for this subcritical binary process; see e.g. Athreya and Ney [1]. The upper bound $A(p) \leq \frac{1}{2}$ follows because $S_1 = p$ gives $S_1/(2p) = \frac{1}{2}$, while the sequence $S_n/(2p)^n$ is decreasing, since $S_{n+1}/S_n = 2p - pS_n < 2p$; consequently $1/A(p) \geq 2$. The older thesis notation sometimes wrote $S_n \rightarrow A(2P)^n$ with a single letter A , and we standardise here on $A(p)$.)

The limit (5) is well defined and can be read off numerically by iterating (3). Figure 1 plots the reciprocal ratio $(2p)^n/S_n$ against n for several subcritical values of p : each trajectory settles to a horizontal asymptote $1/A(p)$, and that asymptote sits higher the closer p comes to $\frac{1}{2}$ from below.

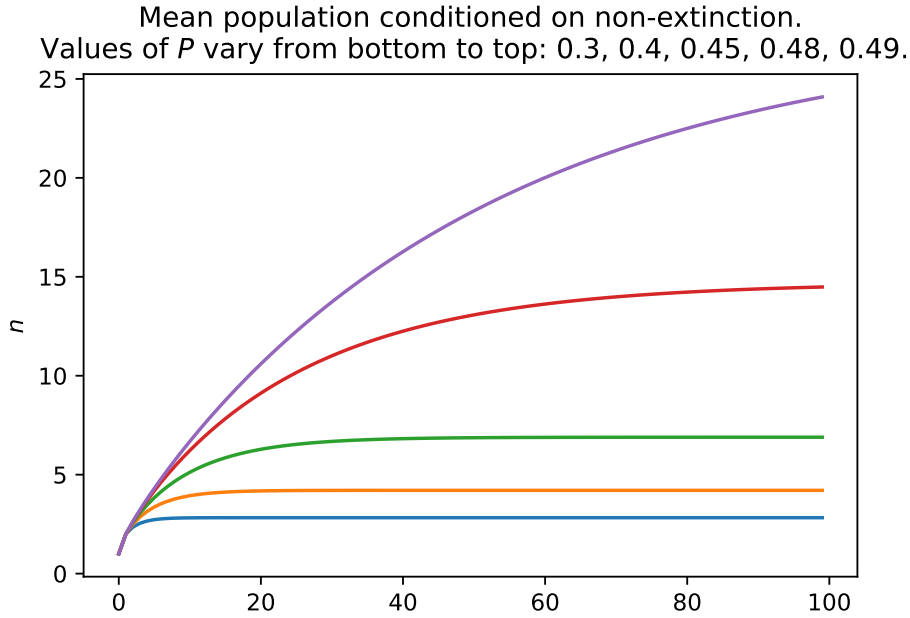


Figure 1: The ratio $(2p)^n/S_n$ against generation n , for subcritical division probabilities p . Curves nearer the top correspond to p closer to criticality (e.g. $p = 0.49$); lower curves use smaller p (e.g. $p = 0.3$). The late-time plateau is $1/A(p)$.

3.4 Quasi-stationarity and conditional mean

When absorption at 0 is certain, one may still ask how the process behaves on the event of non-extinction. A limiting conditional distribution of that kind, when it exists, is called a *quasi-stationary* distribution, and the plateau just seen in Figure 1 is the first sign of one.

Its mean is elementary once (4) is granted. Decomposing the unconditional mean over the extinction event,

$$\mathbb{E}(\mathbf{Z}_n) = \Pr\{\mathbf{Z}_n \neq 0\} \mathbb{E}(\mathbf{Z}_n \mid \mathbf{Z}_n \neq 0),$$

gives

$$\mathbb{E}(\mathbf{Z}_n \mid \mathbf{Z}_n \neq 0) = \frac{(2p)^n}{S_n},$$

which is precisely the ratio plotted above. Sending $n \rightarrow \infty$ and using (4),

$$\lim_{n \rightarrow \infty} \mathbb{E}(\mathbf{Z}_n \mid \mathbf{Z}_n \neq 0) = \frac{1}{A(p)}. \quad (6)$$

In the subcritical death-or-division process, then, lineages that are still alive at large n have expected size tending to the constant $1/A(p)$ —a size scale set by the division probability alone.

3.5 Conditional second moment and variance

A stable conditional mean need not imply a stable spread, so it is worth pushing the same calculation one moment further. For $p < \frac{1}{2}$ the second moment is

$$\mathbb{E}(\mathbf{Z}_n^2) = (2p)^n \left(1 + \frac{1}{1-2p} \right) - (2p)^{2n} \left(\frac{1}{1-2p} \right), \quad (7)$$

the derivation—via the moment generating function and a product rule along the iterate of ϕ —being recorded in the chapter appendix. Conditioning as before and letting $n \rightarrow \infty$,

$$\mathbb{E}(\mathbf{Z}_n^2 \mid \mathbf{Z}_n \neq 0) \rightarrow \frac{1}{A(p)} \left(1 + \frac{1}{1-2p} \right), \quad (8)$$

$$\text{Var}(\mathbf{Z}_n \mid \mathbf{Z}_n \neq 0) \rightarrow \frac{1}{A(p)} \left(1 + \frac{1}{1-2p} \right) - \frac{1}{A(p)^2}. \quad (9)$$

Both the conditional mean and the conditional variance therefore approach finite limits, so the late-time picture on survival is not merely a fixed mean size but a genuine quasi-stationary fluctuation scale.

3.6 What is elementary and what is not

The constant $A(p)$ is as concrete as a limit of an explicit recurrence: it can be computed to any precision by iterating (3) and forming (5). Concreteness of that kind is not the same as a closed form, and the contrast is sharp. The continuous-time birth–death process treated in the next section admits a fully closed elementary expression for the analogous reciprocal conditional mean, written directly in the rate parameters; no comparably elementary closed form for $p \mapsto A(p)$ on $p \in [0, \frac{1}{2})$ is available. The obstruction, and the usable representations that survive it, are the subject of Path C. For the rest of Path A it is enough to treat $A(p)$ as a well-defined, numerically accessible factor that converts a decaying survival probability into a stable conditional mean $1/A(p)$.

4 Continuous-time birth–death and quasi-stationarity

We turn to the linear birth–death process in continuous time. The same subcritical phenomena appear here as in discrete time—certain extinction, together with a non-trivial conditional mean given survival—but now the reciprocal of that conditional mean is elementary in the rates. That contrast with the discrete $A(p)$ is the main structural point of the section. Along the way we record the master-equation bridge to a probability generating function partial differential equation, so that the method of characteristics later in the chapter is not a cold open.

4.1 Process and mean ODE

Let $\mathbf{X}_t$ be a continuous-time Markov chain on $\{0, 1, 2, \dots\}$ with per capita birth rate $\beta \geq 0$ and death rate $\mu \geq 0$, so that for $n \geq 1$ and small $\Delta t > 0$,

$$\begin{aligned}\Pr\{\mathbf{X}_{t+\Delta t} = n+1 \mid \mathbf{X}_t = n\} &= \beta n \Delta t + o(\Delta t), \\ \Pr\{\mathbf{X}_{t+\Delta t} = n-1 \mid \mathbf{X}_t = n\} &= \mu n \Delta t + o(\Delta t),\end{aligned}$$

with 0 absorbing. Write $N = \mathbf{X}_0$ for the initial count, taken deterministic. Because the rates are linear in n , the unconditional mean $M(t) = \mathbb{E}(\mathbf{X}_t)$ closes on itself and satisfies

$$\frac{dM}{dt} = (\beta - \mu)M, \quad M(0) = N, \quad (10)$$

hence $M(t) = N e^{(\beta - \mu)t}$ whenever the mean exists. The process is subcritical, critical, or supercritical according as $\beta - \mu$ is negative, zero, or positive.

4.2 Survival probability

The extinction probability $p_0(t) = \Pr\{\mathbf{X}_t = 0\}$ for the linear birth–death process is classical. With initial condition $\mathbf{X}_0 = N$,

$$p_0(t) = \begin{cases} \left(\frac{\mu - \mu e^{(\mu - \beta)t}}{\beta - \mu e^{(\mu - \beta)t}} \right)^N, & \beta \neq \mu, \\ \left(\frac{\beta t}{1 + \beta t} \right)^N, & \beta = \mu. \end{cases} \quad (11)$$

(The formula is written here with birth rate β in place of the λ of many textbooks.) The survival probability is then $S^{(N)}(t) = 1 - p_0(t)$, and the N th power in (11) records that the N founding lineages must all die out independently.

4.3 Limiting conditional mean

On the event of non-extinction,

$$\mathbb{E}(\mathbf{X}_t \mid \mathbf{X}_t > 0) = \frac{\mathbb{E}(\mathbf{X}_t)}{S^{(N)}(t)}. \quad (12)$$

In the *subcritical* regime $\beta < \mu$ both numerator and denominator tend to zero, and L'Hôpital's rule—or, equivalently, direct asymptotics of (11)—yields a finite positive limit, which under standard linear rates does not depend on the initial count N :

$$\lim_{t \rightarrow \infty} \mathbb{E}(\mathbf{X}_t \mid \mathbf{X}_t > 0) = \frac{\mu}{\mu - \beta} = \frac{1}{1 - \beta/\mu}. \quad (13)$$

The quasi-stationary mean size is thus an explicit rational function of the two rates, and it depends on them only through the ratio β/μ .

For a single ancestor ($N = 1$) and $\beta \neq \mu$, equations (11)–(12) likewise give a closed conditional mean at finite time; its large- t limit is again (13) when $\beta < \mu$.

4.4 Continuous-time analogue of $A(p)$

In discrete time the reciprocal of the limiting conditional mean was $A(p)$. The counterpart suggested by (13) is therefore the dimensionless factor

$$A_{\text{ct}}(\beta, \mu) := \frac{\mu - \beta}{\mu} = 1 - \frac{\beta}{\mu}, \quad \beta < \mu, \quad (14)$$

so that

$$\lim_{t \rightarrow \infty} \mathbb{E}(\mathbf{X}_t \mid \mathbf{X}_t > 0) = \frac{1}{A_{\text{ct}}(\beta, \mu)}.$$

(The notation A_{ct} is used here for the reciprocal limiting conditional mean in the subcritical regime, independent of N for linear rates.)

A comparison with discrete $A(p)$ requires the two models to be put on one axis, and the natural continuum matching of rates to a division probability is

$$p := \frac{\beta}{\beta + \mu}. \quad (15)$$

Then $\beta/\mu = p/(1-p)$ and

$$A_{\text{ct}} = \frac{1-2p}{1-p}, \quad p \in [0, \tfrac{1}{2}). \quad (16)$$

At $p = 0$ one has $A_{\text{ct}} = 1$; as $p \rightarrow \frac{1}{2}^-$, $A_{\text{ct}} \rightarrow 0^+$ and the conditional mean diverges. Both endpoints are therefore under control for the continuous formula.

Figure 2 sets A_{ct} , read as a function of p through (16), against numerical values of the discrete $A(p)$. The two curves have the same shape but not the same normalisation: as noted in Section 3.3, discrete $A(p)$ lies in $(0, \frac{1}{2}]$ and so approaches $\frac{1}{2}$ rather than 1 as $p \rightarrow 0$.

Core contrast (stated, not proved). The continuous subcritical model supplies an elementary closed form for the reciprocal quasi-stationary mean. The discrete factor $A(p)$ is equally well defined as a limit and equally usable in formulae such as (6), yet it is not given by a comparably elementary closed expression on $p \in [0, \frac{1}{2})$. Path C returns to that obstruction; Path A needs only the distinction.

4.5 Near criticality

As β approaches μ from below, $A_{\text{ct}} \rightarrow 0$ and the conditional mean (13) becomes arbitrarily large. A process nearly indifferent between gain and loss can therefore, on the rare event of long survival, carry a large typical size without ever having a positive probability of eternal persistence. The same divergence of $1/A(p)$ appears in discrete time as $p \rightarrow \frac{1}{2}^-$. The point is interpretive rather than a new theorem, but it is the one worth carrying forward: criticality organises not only extinction probabilities but the scale of what survival looks like when it occurs.

4.6 From the master equation to a probability generating function PDE

Let $p_n(t) = \Pr \{\mathbf{X}_t = n\}$ and $G(z, t) = \sum_{n \geq 0} p_n(t) z^n = \mathbb{E}(z^{\mathbf{X}_t})$. Substituting the master equation of Section 2.5, with rates β, μ , and rearranging yields the first-order equation

$$\frac{\partial G}{\partial t} = (\beta z^2 - (\beta + \mu)z + \mu) \frac{\partial G}{\partial z}, \quad (17)$$

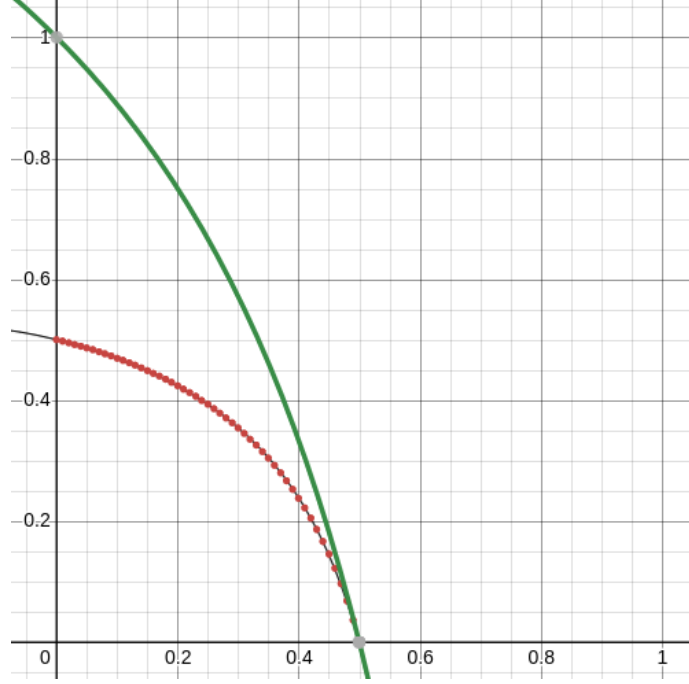


Figure 2: Continuous factor $A_{\text{ct}} = (1-2p)/(1-p)$ (green) against discrete $A(p)$ estimated numerically by iterating (3) (red). Horizontal axis: effective division probability p . (Any additional curve in the source graphic is illustrative only and is not used as a cited closed form.)

for $z \in [0, 1]$, with initial condition $G(z, 0) = z^N$ when $\mathbf{X}_0 = N$.

Equation (17) is a linear transport equation in (z, t) , and it is solved by following the curves along which G does not change. Writing $a(z) = \beta z^2 - (\beta + \mu)z + \mu$, so that the PDE reads $G_t = a(z)G_z$, we have along any curve $z(t)$

$$\frac{d}{dt}G(z(t), t) = (a(z) + \dot{z})G_z.$$

Hence G is constant on characteristics precisely when

$$\frac{dz}{dt} = -a(z) = -(\beta z^2 - (\beta + \mu)z + \mu) = (\beta z - \mu)(1 - z). \quad (18)$$

(The pure-death case $\beta = 0$ gives $\dot{z} = \mu(z - 1)$, matching the explicit solution $G(z, t) = (1 - e^{-\mu t} + ze^{-\mu t})^N$.)

Solving (18) with $G(z, 0) = z^N$ recovers the classical closed-form generating function of the linear birth–death process, standard in the textbooks; the single-ancestor case determines the general N case by independence of lineages. In particular $p_0(t) = G(0, t)$ reproduces (11), the formula already used above for survival and conditional means. That closes the loop, and it is the verification the bridge requires: the PDE, the characteristic field (18), and consistency with the extinction formula.

Exit contract for the method of characteristics. The later method-of-characteristics development may therefore assume three things: (i) that a univariate CTMC probability generating function satisfies a first-order PDE obtained from the master equation; (ii) that for linear birth–death this PDE is (17), the characteristics are (18), and the solution is the classical closed form with $p_0 = G(0, \cdot)$ as in (11); and (iii) that the same idea extends when the state is multivariate or when

further transitions modify the transport field. The method-of-characteristics section develops the absorption ladder (with and without birth and death) in detail; catastrophe rates in full BDC models are taken up in later chapters, reusing the same generating-function pattern.

5 Survivability of small populations

The previous sections fixed the mean growth regime and the quasi-stationary picture on survival. Here the question is different, and more directly applied: how does the *initial* count change the chance that a branching population is still alive at a large generation? The answer, in short, is that near criticality survival probabilities are highly sensitive to the initial cohort size k —and that sensitivity is a principal reason why stochastic branching, and later birth–death–catastrophe models, matter for early infection at all.

5.1 Initial cohort of size k

Return to the discrete death-or-division process with division probability p , and write ${}_1u_n$ for the extinction probability by generation n starting from a single individual, so that ${}_1S_n = 1 - {}_1u_n$. If the process instead starts from k independent founders, extinction by generation n requires every lineage to have died out:

$$\Pr\{\text{extinction by } n \mid \mathbf{Z}_0 = k\} = ({}_1u_n)^k.$$

Hence the survival probability is

$$S_n^{(k)} = 1 - ({}_1u_n)^k, \tag{19}$$

which for $k > 1$ is strictly larger than ${}_1S_n$. The size of the gain is easiest to read as a ratio,

$$\frac{S_n^{(k)}}{{}_1S_n} = \sum_{j=0}^{k-1} ({}_1u_n)^j \leq k,$$

and this ratio approaches its ceiling k whenever ${}_1u_n \rightarrow 1$ —that is, at late times in the subcritical regime, or wherever survival decays slowly near criticality. Founders then contribute almost independently, and the gain is close to proportional in k .

Figures 3 and 4 show $S_n^{(k)}$ through 500 generations. Raising k lifts the whole survival curve, and the effect is itself highly sensitive to criticality: it strengthens as p approaches $\frac{1}{2}$ and remains strong just into the supercritical regime. Continuous time carries the same qualitative message, and for the same reason. For independent linear birth–death lineages the extinction probability with N founders is $(p_0^{(1)}(t))^N$, so survival is $1 - (p_0^{(1)}(t))^N$, again monotone in the founding count and again sharply dependent on it near criticality.

5.2 Reading for early infection

Suppose an infection is only slightly supercritical, or spends an early window near criticality. A modest increase in the founding inoculum can then change the odds of persistence far more than a naive proportional reading of the mean would suggest. Mechanisms that raise the effective early count—a sheltered intracellular burst before extracellular exposure, for example—are therefore natural candidates for selective advantage even when the later per capita rates are entirely unchanged. The present section records only the branching-process mechanism behind that reading; the pathogen life histories in which it is realised are taken up in later chapters.

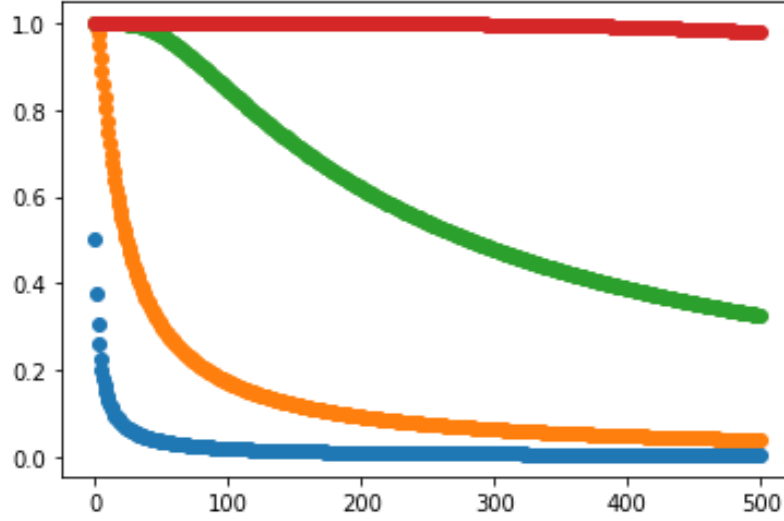


Figure 3: Survival probability out to 500 generations for initial counts $k \in \{1, 10, 100, 1000\}$ (blue through red), at criticality $p = \frac{1}{2}$.

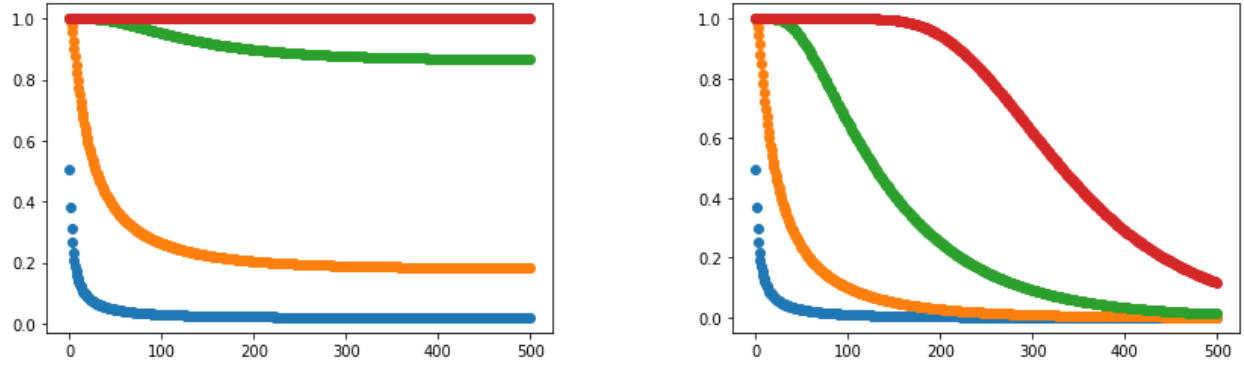


Figure 4: As in Figure 3, but slightly supercritical ($p = 0.505$, left) and slightly subcritical ($p = 0.495$, right). The gain from larger k is strongest near $p = \frac{1}{2}$.

5.3 Teaser: discrete birth–death–catastrophe

A discrete-time birth–death–catastrophe (BDC) process adds, at each generation, a chance that the whole population is removed by a catastrophe event such as a rupture, on top of the ordinary death and division already in place.

Suppose each individual independently triggers catastrophe with probability κ when the count is n , so that the probability of no catastrophe in that generation is $(1 - \kappa)^n$. Because this survival factor depends on the current size, surviving paths are reweighted by a quantity that depends on the whole trajectory of sizes rather than on its endpoint. The law conditional on avoiding catastrophe is consequently *not* the plain Galton–Watson law of Section 3.4, and the conditional mean is not simply $1/A(p)$.

None of that takes the model out of reach. One can still read the construction as a size-dependent thinning of the offspring mechanism: the process remains in the branching family, and quasi-stationary ideas continue to apply, though with constants that now depend on κ . In a pure

birth–catastrophe caricature with no death, counts are deterministic between ruptures and several probabilities are elementary; restoring death makes them random again. The full continuous-time, rate-based bookkeeping is the subject of the birth–death–catastrophe chapters. For Path A it is enough that small- k survival and carefully conditioned quasi-stationarity remain the organising motifs once catastrophes are present.

6 Checkpoint: the working toolkit

The sections above form Path A of the chapter: the shared vocabulary and results used throughout the thesis. Before the longer technical developments begin, it is worth collecting what has now been fixed.

Markov structure. Discrete- and continuous-time Markov chains, absorbing states, rate descriptions for linear birth and death, and the master equation as a balance of probability flux.

Generating functions. The probability generating function as a transform of a non-negative integer law, together with its two working roles: functional iteration in discrete branching, and, through the master equation, a first-order PDE for linear birth–death.

Criticality. Subcritical, critical, and supercritical regimes in discrete time (through $m = 2p$, or a general offspring mean) and in continuous time (through $\beta - \mu$); extinction versus survival; and the way criticality organises quasi-stationary scales near the threshold.

Quasi-stationarity and $A(p)$. In subcritical death-or-division branching, $S_n \sim A(p)(2p)^n$ and $\mathbb{E}(\mathbf{Z}_n \mid \mathbf{Z}_n \neq 0) \rightarrow 1/A(p)$, with a finite limiting conditional variance alongside. In subcritical continuous birth–death, the reciprocal conditional mean is the elementary factor $A_{\text{ct}} = (1 - 2p)/(1 - p)$ under the rate matching $p = \beta/(\beta + \mu)$. The discrete factor is well defined and computable, but not elementary in that same sense; Path C returns to the fact and to what lies behind it.

Small populations. Survival probabilities with k founders, $S_n^{(k)} = 1 - (1u_n)^k$, are highly sensitive to k near criticality. That sensitivity is what motivates a careful stochastic treatment of early infection and of small-count regimes generally.

How to read what follows.

- **Path B (method of characteristics).** Starts from the univariate probability generating function PDE of Section 4.6 and develops the analytical ladder for continuous-time absorption models (with and without birth and death). Required reading for anyone who will use those partial differential equations; full catastrophe structure appears in the later BDC chapters.
- **Path C ($A(p)$ and Koenigs).** Optional depth: representations, numerics, and the closed-form obstruction for discrete $A(p)$. It can be deferred whenever the core contrast above is enough for the application at hand.

The appendix collects the heavy derivations, labelled by path—discrete moments, long characteristic calculations, Koenigs detail. Forward references to later thesis chapters are deliberately sparse, since the toolkit is meant to travel without a dependency table.

7 Method of characteristics

Path B develops the analytical ladder for probability generating functions of continuous-time absorption models. Section 4.6 already recorded the univariate linear birth–death case: the master equation compresses to a first-order PDE, solved along characteristics. Here the state is bivariate—intracellular and extracellular counts—and the transitions are absorption, death, and birth. The same idea applies: write the forward equations, pass to a generating function $G(x, y, t)$, obtain a first-order PDE, and integrate along characteristics.

The original motivation is multi-host pathogenesis: once a cell ruptures, a large extracellular cohort becomes available for absorption by the remaining hosts. As a step towards that picture one studies single-host “absorption models”, in which a pathogenic cohort moves from the medium into one cell, with or without intracellular birth and death. Immediate whole-cohort transfer between cells is a harder assumption to defend; absorption over an intermediate period is more realistic. The calculations below supply the generating-function technology for those building blocks.

7.1 Only absorption

The simplest absorption process is a single host cell that takes up pathogenic particles but in which no subsequent dynamics occur — neither birth nor death of pathogen within the host.

Let $\mathbf{X}_t$ and $\mathbf{Y}_t$ denote the number of pathogenic particles within and outside the host cell respectively; the same labelling is used in the later versions of the single-host model. Events are assumed throughout to be stochastic, separated by exponentially distributed inter-event times, a property that follows from taking event rates to be time-independent. Our continuous-time stochastic process $\{(\mathbf{X}_t, \mathbf{Y}_t) : t \in [0, \infty)\}$ therefore conforms to the memoryless Markov property.

With only absorption to consider, the model is specified by the following probabilities — outcome-likelihoods over an infinitesimal duration, Δt (taken in the limit to 0).

$$\begin{aligned} \Pr\{\Delta \mathbf{X} = 1, \Delta \mathbf{Y} = -1 \mid \mathbf{Y}_t = j\} &= j\alpha\Delta t, & (\text{absorption}) \\ \Pr\{\Delta \mathbf{X} = 0, \Delta \mathbf{Y} = 0 \mid \mathbf{Y}_t = j\} &= 1 - j\alpha\Delta t, & (\text{no event}) \end{aligned}$$

Here $\Delta \mathbf{X} = \mathbf{X}_{t+\Delta t} - \mathbf{X}_t$ (and likewise for $\mathbf{Y}$). Notice the multiplication by j , the number of extracellular particles: each unit is subject to the same probability, $\alpha\Delta t$, of being taken up.

The initial condition is fixed as $\mathbf{X}_0 = 0$, $\mathbf{Y}_0 = N$ — a starter population of census N occupying the extracellular medium.

Defining state probabilities, $p_{i,j}(t) = \Pr\{\mathbf{X}_t = i, \mathbf{Y}_t = j\}$, we can write down the forward Kolmogorov equations:

$$\frac{dp_{i,j}}{dt} = \alpha(j+1)p_{i-1,j+1} - \alpha jp_{i,j}. \quad (20)$$

Then, with the probability generating function defined,

$$G(x, y, t) = \sum_i \sum_j p_{i,j}(t) x^i y^j \quad (21)$$

($= \mathbb{E}[x^{\mathbf{X}_t} y^{\mathbf{Y}_t}]$), we can derive a PDE for $G(x, y, t)$ in the following way.

First, multiply each equation in (20) by $x^i y^j$,

$$\frac{\partial p_{i,j} x^i y^j}{\partial t} = \alpha(j+1)p_{i-1,j+1} x^i y^j - \alpha jp_{i,j} x^i y^j.$$

Then sum over i and j for non-zero $p_{i,j}(t)$,

$$\sum_{i=0}^N \sum_{j=0}^N \frac{\partial p_{i,j} x^i y^j}{\partial t} = \alpha \sum_{i=1}^{N+1} \sum_{j=-1}^{N-1} (j+1) p_{i-1,j+1} x^i y^j - \alpha \sum_{i=0}^N \sum_{j=0}^N j p_{i,j} x^i y^j. \quad (22)$$

(Note, for any $i, j > N$, and when $i + j \neq N$, $p_{i,j}(t) = 0 \forall t$. This is because pathogenic particles in this model are neither created nor destroyed.)

Rearranging (22), we find

$$\sum_{i=0}^N \sum_{j=0}^N \frac{\partial p_{i,j} x^i y^j}{\partial t} = \alpha \left(x \sum_{i=0}^N \sum_{j=0}^N j p_{i,j} x^i y^{j-1} - y \sum_{i=0}^N \sum_{j=0}^N j p_{i,j} x^i y^{j-1} \right),$$

which can be written in terms of the following partial derivatives,

$$\sum_{i=0}^N \sum_{j=0}^N \frac{\partial p_{i,j} x^i y^j}{\partial t} = \alpha (x - y) \sum_{i=0}^N \sum_{j=0}^N \frac{\partial p_{i,j} x^i y^j}{\partial y}.$$

Notice, with (21),

$$\frac{\partial G(x, y, t)}{\partial t} = \alpha (x - y) \frac{\partial G(x, y, t)}{\partial y}; \quad (23)$$

the chosen initial condition corresponding to $G(x, y, 0) = y^N$.

This first-order linear initial value problem, it turns out, proves quite difficult to solve. The method of characteristics, though practicable by definition on equations of three variables, is not elementary by inspection here. As it happens, though, we can obtain the solution by other means.

Benefits of the independence property

The function $G(x, y, t)$ referred to above was specific to the initial condition $\mathbf{X}_0 = 0$, $\mathbf{Y}_0 = N$. It helps to carry that starting state explicitly in the notation,

$$G_n(x, y, t) = \sum_i \sum_j \Pr \{ \mathbf{X}_t = i, \mathbf{Y}_t = j \mid \mathbf{X}_0 = 0, \mathbf{Y}_0 = n \} x^i y^j.$$

In a process of absorption alone, each of the N pathogenic particles gets taken up by the host cell one by one, and a single particle of the cohort behaves without reference to the movements of its neighbours. Because of this independence, our variables $\mathbf{X}_t$ & $\mathbf{Y}_t$ can be thought of as the sum of N sub-variables, $\mathbf{X}_t^{(k)}$ & $\mathbf{Y}_t^{(k)}$, each pertaining only to the movements of a single pathogenic unit:

$$\begin{aligned} \mathbf{X}_t &= \mathbf{X}_t^{(1)} + \mathbf{X}_t^{(2)} + \dots + \mathbf{X}_t^{(n)} \\ \mathbf{Y}_t &= \mathbf{Y}_t^{(1)} + \mathbf{Y}_t^{(2)} + \dots + \mathbf{Y}_t^{(n)} \end{aligned}$$

This observation allows us to prove that

$$G_n(x, y, t) = (G_1(x, y, t))^n : \quad (24)$$

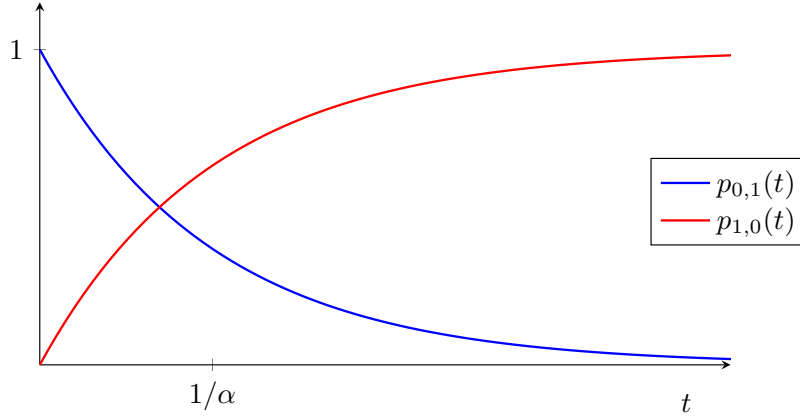
$$G_n(x, y, t) = \mathbb{E} \left(x^{\sum_{k=1}^n \mathbf{X}_t^{(k)}} y^{\sum_{k=1}^n \mathbf{Y}_t^{(k)}} \right)$$

$$\begin{aligned}
&= \mathbb{E} \left(\prod_{k=1}^n x^{\mathbf{X}_t^{(k)}} \prod_{k=1}^n y^{\mathbf{Y}_t^{(k)}} \right) \\
&= \prod_{k=1}^n \mathbb{E} \left(x^{\mathbf{X}_t^{(k)}} y^{\mathbf{Y}_t^{(k)}} \right) \\
&= \prod_{k=1}^n G_1(x, y, t) \\
&= (G_1(x, y, t))^n
\end{aligned}$$

Now consider a system with only one pathogenic particle to begin with; $\mathbf{X}_0 = 0$, $\mathbf{Y}_0 = 1$. It would after some time be absorbed, and this would be the final state of the system. No other states could be reached — quite simply: $p_{i,j}(t) = 0 \ \forall t$ for $i + j \neq 1$.

The state probabilities of this simple case can hence be fully determined from the forward Kolmogorov equations.

$$\begin{aligned}
\frac{dp_{0,1}(t)}{dt} &= -\alpha p_{0,1}(t) & \frac{dp_{1,0}(t)}{dt} &= \alpha p_{0,1}(t) \\
\Rightarrow & & & \\
p_{0,1}(t) &= e^{-\alpha t} & p_{1,0}(t) &= 1 - e^{-\alpha t}
\end{aligned}$$



Given that these are the only two non-zero state probabilities of the single-particle system, we can write out the associated probability generating function as

$$\begin{aligned}
G_1(x, y, t) &= p_{1,0}(t)x + p_{0,1}(t)y \\
&= (1 - e^{-\alpha t})x + e^{-\alpha t}y.
\end{aligned}$$

Applying relation (24) then defines the probability generating function for this absorption-only system generally, for any initial state,

$$G_n(x, y, t) = \left((1 - e^{-\alpha t})x + e^{-\alpha t}y \right)^n;$$

which does satisfy equation (23) along with its initial condition,

$$G_n(x, y, 0) = y^n.$$

7.2 Absorption-death

In reality, pathogenic particles resident in a host cell may be subject to death.

Setting up a model as above, with $\mathbf{X}_t$ & $\mathbf{Y}_t$ representing the count of intracellular and extracellular pathogen, we can incorporate particle death with the following addition.

$$\begin{aligned} \Pr \{ \Delta \mathbf{X} = 1, \Delta \mathbf{Y} = -1 \mid \mathbf{Y}_t = j \} &= j\alpha \Delta t, & (\text{absorption}) \\ \Pr \{ \Delta \mathbf{X} = -1, \Delta \mathbf{Y} = 0 \mid \mathbf{X}_t = i \} &= i\mu \Delta t, & (\text{death}) \\ \Pr \{ \Delta \mathbf{X} = 0, \Delta \mathbf{Y} = 0 \mid \mathbf{Y}_t = j, \mathbf{X}_t = i \} &= 1 - (j\alpha + i\mu) \Delta t, & (\text{no event}) \end{aligned}$$

From here an analogous line of reasoning can be followed, and a familiar difficulty is met on arrival at the first order linear PDE,

$$\frac{\partial G(x, y, t)}{\partial t} = \mu(1 - x) \frac{\partial G(x, y, t)}{\partial x} + \alpha(x - y) \frac{\partial G(x, y, t)}{\partial y}.$$

This point being reached from a similar derivation using the forward Kolmogorov equations:

$$\frac{dp_{i,j}}{dt} = \alpha(j+1)p_{i-1,j+1} + \mu(i+1)p_{i+1,j} - (\alpha j + \mu i)p_{i,j}.$$

But as above, the solution can be constructed with a work-around employing the relation

$$G_n(x, y, t) = (G_1(x, y, t))^n.$$

As before, individual particles are independent from one another in their movements.

Without birth events, the maximum number of particles in the system is restricted to its initial value. So, when considering the case of an isolated pathogenic unit, the probability generating function takes its form,

$$G_1(x, y, t) = p_{0,0} + p_{1,0}x + p_{0,1}y.$$

These state-probabilities can be determined as follows using the forward Kolmogorov equations.

$$\begin{aligned} \frac{dp_{0,1}(t)}{dt} &= -\alpha p_{0,1}(t) & \frac{dp_{1,0}(t)}{dt} &= \alpha p_{0,1}(t) - \mu p_{1,0}(t) \\ \Rightarrow & & & \\ p_{0,1}(t) &= e^{-\alpha t} & p_{1,0}(t) &= \frac{\alpha}{\mu - \alpha} (e^{-\alpha t} - e^{-\mu t}) & p_{0,0}(t) &= 1 - p_{0,1}(t) - p_{1,0}(t) \end{aligned}$$

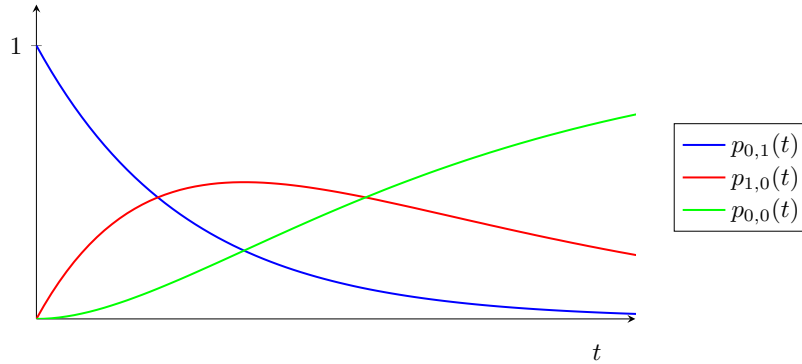


Figure 5: Characteristic geometry for the absorption-death model.

Relation (24) then leads, as above, to the general probability generating function, $G_n(x, y, t)$:

$$G_n(x, y, t) = \left(1 - \frac{\mu}{\mu - \alpha} e^{-\alpha t} + \frac{\alpha}{\mu - \alpha} e^{-\mu t} + \frac{\alpha}{\mu - \alpha} (e^{-\alpha t} - e^{-\mu t}) x + e^{-\alpha t} y\right)^n$$

7.3 Method of characteristics (absorption-death)

Before continuing to the case of absorption-birth-death, where replication may occur within the host cell, let us revisit absorption-death from the point of its PDE for $G(x, y, t)$.

$$\frac{\partial G(x, y, t)}{\partial t} = \mu(1 - x) \frac{\partial G(x, y, t)}{\partial x} + \alpha(x - y) \frac{\partial G(x, y, t)}{\partial y}, \quad (25)$$

with initial condition $G(x, y, 0) = y^N$.

The method of characteristics is the standard tool for first-order linear PDEs of this type for 3 variate problems.

We begin by introducing a change of variables $(x, y, t) \rightarrow (s, r, \tau)$:

$$G(x, y, t) = G(x(s, r, \tau), y(s, r, \tau), t(s, r, \tau)) \equiv G(s, r, \tau)$$

The method works by seeking solutions on the characteristic surfaces of constant s and r ; $G(\tau)$. To do so, we must introduce an initial surface in the s, r, τ space:

$$x(s, r, 0) = s \quad y(s, r, 0) = r \quad t(s, r, 0) = 0 \quad (26)$$

With the probability generating function now a function of only one variable, τ , we can write down the characteristic equations as ODEs. First, consider,

$$\frac{dG}{d\tau} = \frac{\partial G}{\partial t} \frac{dt}{d\tau} + \frac{\partial G}{\partial x} \frac{dx}{d\tau} + \frac{\partial G}{\partial y} \frac{dy}{d\tau}.$$

Notice similarity with (7.2),

$$0 = -\frac{\partial G}{\partial t} + \mu(1 - x) \frac{\partial G}{\partial x} + \alpha(x - y) \frac{\partial G}{\partial y}.$$

Comparing coefficients, we can write down the characteristic ODEs:

$$\frac{dG}{d\tau} = 0 \quad (27)$$

$$\frac{dt}{d\tau} = -1 \quad (28)$$

$$\frac{dx}{d\tau} = \mu(1 - x) \quad (29)$$

$$\frac{dy}{d\tau} = \alpha(x - y) \quad (30)$$

(27)

Solving the first characteristic equation,

$$\frac{dG}{d\tau} = 0$$

$$G = \phi(s, r)$$

ϕ constant on surfaces of constant s , and r .

With (26), the initial condition on G translates to $G(s, r, \tau = 0) = r^N$, which informs us that $\phi(s, r) = r^N$. Hence,

$$G(s, r, \tau) = r^N. \quad (31)$$

(28)

$$\begin{aligned} \frac{dt}{d\tau} &= -1 \\ t &= -\tau + c(s, r) \end{aligned}$$

Using (26), we can fix the constant, $c(s, r) = 0$.

$$t = -\tau \quad (32)$$

(29)

$$\begin{aligned} \frac{dx}{d\tau} &= \mu(1 - x) \\ &\vdots \\ 1 - x &= c(s, r)e^{-\mu\tau}. \end{aligned}$$

With initial condition, (26), $c(s, r) = 1 - s$;

$$1 - x = (1 - s)e^{-\mu\tau}. \quad (33)$$

(30)

$$\begin{aligned} \frac{dy}{d\tau} &= \alpha(x - y) \\ \frac{dy}{d\tau} &= \alpha(1 - (1 - s)e^{-\mu\tau} - y) \\ &\vdots \\ y &= 1 - \frac{\alpha}{\alpha - \mu}(1 - s)e^{-\mu\tau} + c(s, r)e^{-\alpha\tau}. \end{aligned}$$

Applying (26), $c(s, r) = r + \frac{\alpha}{\alpha - \mu}(1 - s) - 1$;

$$y = 1 - \frac{\alpha}{\alpha - \mu}(1 - s)e^{-\mu\tau} + \left(r + \frac{\alpha}{\alpha - \mu}(1 - s) - 1\right)e^{-\alpha\tau}. \quad (34)$$

Finally, if we rearrange (34) for r , also substituting $s(x, y, t)$ from (33), and $\tau = -t$ from (32), we find:

$$r = 1 - \frac{\mu}{\mu - \alpha}e^{-\alpha t} + \frac{\alpha}{\mu - \alpha}e^{-\mu t} + \frac{\alpha}{\mu - \alpha}(e^{-\alpha t} - e^{-\mu t})x + e^{-\alpha t}y$$

Then, with $G = r^N$ ((31)), this matches the solution obtained in the previous section,

$$G(x, y, t) = \left(1 - \frac{\mu}{\mu - \alpha}e^{-\alpha t} + \frac{\alpha}{\mu - \alpha}e^{-\mu t} + \frac{\alpha}{\mu - \alpha}(e^{-\alpha t} - e^{-\mu t})x + e^{-\alpha t}y\right)^N.$$

7.4 Absorption-birth-death

Introducing replication for particles inside the host cell, we will see that though the independence property is still present, it no longer serves as a work-around in finding the probability generating function $G(x, y, t)$. This is because, given a single initial infectious particle, with birth events an infinite number of states can be reached; hence $G_1(x, y, t)$ will be constituted by an infinite number of terms.

We are now equipped, however, to solve the PDE for $G(x, y, t)$ directly, and this is the approach we will take.

Setting up the model like in previous cases, $\mathbf{X}_t$ & $\mathbf{Y}_t$ represent the counts of intracellular and extracellular pathogenic particles. Birth events are added to the transition rates as follows.

$$\begin{aligned} \Pr\{\Delta\mathbf{X} = 1, \Delta\mathbf{Y} = -1 \mid \mathbf{Y}_t = j\} &= j\alpha\Delta t, & (\text{absorption}) \\ \Pr\{\Delta\mathbf{X} = 1, \Delta\mathbf{Y} = 0 \mid \mathbf{X}_t = i\} &= i\beta\Delta t, & (\text{birth}) \\ \Pr\{\Delta\mathbf{X} = -1, \Delta\mathbf{Y} = 0 \mid \mathbf{X}_t = i\} &= i\mu\Delta t, & (\text{death}) \\ \Pr\{\Delta\mathbf{X} = 0, \Delta\mathbf{Y} = 0 \mid \mathbf{Y}_t = j, \mathbf{X}_t = i\} &= 1 - (j\alpha + i\mu + i\beta)\Delta t, & (\text{no event}) \end{aligned}$$

The forward Kolmogorov equation is written as such.

$$\frac{\partial p_{i,j}}{\partial t} = \alpha(j+1)p_{i-1,j+1} + \mu(i+1)p_{i+1,j} + \beta(i-1)p_{i-1,j} - (\alpha j + \mu i + \beta i)p_{i,j}$$

From here, again with the reasoning seen above, we can derive the associated PDE for $G(x, y, t)$.

$$\frac{\partial G}{\partial t} = \left(\mu - (\mu + \beta)x + \beta x^2 \right) \frac{\partial G}{\partial x} + \alpha(x - y) \frac{\partial G}{\partial y} \quad (35)$$

To solve this by the method of characteristics we again re-parametrise (x, y, t) as (s, r, τ) , then seek solutions on characteristic surfaces of constant s and r . Doing so brings us to the characteristic ODEs:

$$\frac{dG}{d\tau} = 0 \quad (36)$$

$$\frac{dt}{d\tau} = -1 \quad (37)$$

$$\frac{dx}{d\tau} = \mu - (\mu + \beta)x + \beta x^2 \quad (38)$$

$$\frac{dy}{d\tau} = \alpha(x - y) \quad (39)$$

Applying the same initial conditions on (s, r, τ) as above in (26), we can solve these equations as follows.

(36) & (37)

Like in the previous section the first two characteristic equations yield:

$$G(s, r, \tau) = \phi(s, r),$$

and ultimately,

$$G(s, r, \tau) = r^N. \quad (40)$$

Then,

$$t = -\tau.$$

(38)

This equation has the same structural shape as the survival recurrence for S_n in the discrete branching section, the survival probability.

$$\frac{dx}{d\tau} = \beta(x - x_+)(x - x_-)$$

with,

$$x_{\pm} = \frac{\mu + \beta}{2\beta} \pm \sqrt{\left(\frac{\mu + \beta}{2\beta}\right)^2 - \frac{\mu}{\beta}}.$$

The main difference in its solution with the discrete survival recurrence comes with an s when fixing the constant of integration:

$$x(\tau) = \frac{x_+(s - x_-) - x_-(s - x_+)e^{\beta(x_+ - x_-)\tau}}{(s - x_-) - (s - x_+)e^{\beta(x_+ - x_-)\tau}} \quad (41)$$

(39)

The last characteristic equation is

$$\frac{dy}{d\tau} = \alpha(x - y).$$

To solve this we will need to substitute x as found above, (41). With an integrating factor, the equation becomes,

$$\frac{d}{d\tau} (e^{\alpha\tau} y) = \alpha e^{\alpha\tau} x(\tau).$$

Integrating with respect to τ ,

$$e^{\alpha\tau} y = \alpha \int e^{\alpha\tau} \frac{x_+(s - x_-) - x_-(s - x_+)e^{\beta(x_+ - x_-)\tau}}{(s - x_-) - (s - x_+)e^{\beta(x_+ - x_-)\tau}} d\tau.$$

Then splitting the numerator,

$$\begin{aligned} e^{\alpha\tau} y &= \alpha x_+(s - x_-) \int \frac{e^{\alpha\tau}}{(s - x_-) - (s - x_+)e^{\beta(x_+ - x_-)\tau}} d\tau \\ &\quad - \alpha x_-(s - x_+) \int \frac{e^{[\alpha + \beta(x_+ - x_-)]\tau}}{(s - x_-) - (s - x_+)e^{\beta(x_+ - x_-)\tau}} d\tau. \end{aligned} \quad (42)$$

Each of these integrals can be written in the form,

$$I(\tau) = \int \frac{e^{h\tau}}{p + qe^{k\tau}} d\tau.$$

where h, k, p, q are all constants. For example, in the first integral, $h = \alpha$, $k = \beta(x_+ - x_-)$, $p = (s - x_-)$, and $q = -(s - x_+)$. As it happens, we can solve this integral with:

$$I(\tau) = \frac{e^{h\tau}}{ph} {}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{q}{p}e^{k\tau}\right) + \text{const.} \quad (43)$$

${}_2F_1(a, b; c; z)$ being the hypergeometric function, defined,

$${}_2F_1(a, b; c; z) = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n} \frac{z^n}{n!},$$

and $(p)_n$ is the (rising) Pochhammer symbol:

$$(p)_n = \begin{cases} 1 & n = 1 \\ p(p+1) \cdots (p+n-1) & n > 1 \end{cases}$$

See appendix B.1 for a detailed derivation of (43).

Picking up again at (42), the integrals are specified by (43):
(using $b_1 = \beta(x_+ - x_-)$ for convenience)

$$\int \frac{e^{\alpha\tau}}{(s-x_-) - (s-x_+)e^{b_1\tau}} d\tau = \frac{e^{\alpha\tau}}{\alpha(s-x_-)} {}_2F_1\left(1, \frac{\alpha}{b_1}; \frac{\alpha}{b_1} + 1; \frac{(s-x_+)}{(s-x_-)} e^{b_1\tau}\right),$$

and

$$\int \frac{e^{[\alpha+b_1]\tau}}{(s-x_-) - (s-x_+)e^{b_1\tau}} d\tau = \frac{e^{(\alpha+b_1)\tau}}{(\alpha+b_1)(s-x_-)} {}_2F_1\left(1, \frac{\alpha+b_1}{b_1}; \frac{\alpha+b_1}{b_1} + 1; \frac{(s-x_+)}{(s-x_-)} e^{b_1\tau}\right).$$

Writing these formulas as $I_1(s, r, \tau)$ and $I_2(s, r, \tau)$ respectively, and substituting into (42):

$$e^{\alpha\tau}y = \alpha x_+(s-x_-)I_1(s, r, \tau) - \alpha x_-(s-x_+)I_2(s, r, \tau) + \text{const.} \quad (44)$$

From here, our initial condition, $y(s, r, 0) = r$, fixes the constant —

$$\text{const} = r - \alpha x_+(s-x_-)I_1(s, r, 0) + \alpha x_-(s-x_+)I_2(s, r, 0).$$

Then, with a rearrangement of (44),

$$r = e^{\alpha\tau}y - \alpha x_+(s-x_-)I_1(s, r, \tau) + \alpha x_-(s-x_+)I_2(s, r, \tau) + \alpha x_+(s-x_-)I_1(s, r, 0) - \alpha x_-(s-x_+)I_2(s, r, 0)$$

With I_1, I_2 written explicitly,

$$\begin{aligned} r = & e^{\alpha\tau}y \\ & - (x_+e^{\alpha\tau}) {}_2F_1\left(1, \frac{\alpha}{b_1}; \frac{\alpha}{b_1} + 1; \frac{(s-x_+)}{(s-x_-)} e^{b_1\tau}\right) \\ & + \left(\frac{\alpha}{\alpha+b_1}x_-e^{(\alpha+b_1)\tau}\right) {}_2F_1\left(1, \frac{\alpha+b_1}{b_1}; \frac{\alpha+b_1}{b_1} + 1; \frac{(s-x_+)}{(s-x_-)} e^{b_1\tau}\right) \\ & + (x_+) {}_2F_1\left(1, \frac{\alpha}{b_1}; \frac{\alpha}{b_1} + 1; \frac{(s-x_+)}{(s-x_-)}\right) \\ & - \left(\frac{\alpha}{\alpha+b_1}x_-\right) {}_2F_1\left(1, \frac{\alpha+b_1}{b_1}; \frac{\alpha+b_1}{b_1} + 1; \frac{(s-x_+)}{(s-x_-)}\right) \end{aligned}$$

From (41), we have the relation,

$$\left(\frac{s-x_+}{s-x_-}\right) = \left(\frac{x-x_+}{x-x_-}\right) e^{b_1\tau},$$

which, applied to the above along with the switch, $\tau = -t$, yields

$$r =$$

$$\begin{aligned}
& e^{-\alpha t} y \\
& - \left(x_+ e^{-\alpha t} \right) {}_2F_1 \left(1, \frac{\alpha}{b_1}; \frac{\alpha}{b_1} + 1; \frac{(x - x_+)}{(x - x_-)} \right) \\
& + \left(\frac{\alpha}{\alpha + b_1} x_- e^{-(\alpha + b_1)t} \right) {}_2F_1 \left(1, \frac{\alpha + b_1}{b_1}; \frac{\alpha + b_1}{b_1} + 1; \frac{(x - x_+)}{(x - x_-)} \right) \\
& + (x_+) {}_2F_1 \left(1, \frac{\alpha}{b_1}; \frac{\alpha}{b_1} + 1; \frac{(x - x_+)}{(x - x_-)} e^{b_1 t} \right) \\
& - \left(\frac{\alpha}{\alpha + b_1} x_- \right) {}_2F_1 \left(1, \frac{\alpha + b_1}{b_1}; \frac{\alpha + b_1}{b_1} + 1; \frac{(x - x_+)}{(x - x_-)} e^{b_1 t} \right)
\end{aligned}$$

Finally, with (40),

$$\begin{aligned}
G(x, y, t) = & \left[e^{-\alpha t} y \right. \\
& - \left(x_+ e^{-\alpha t} \right) {}_2F_1 \left(1, \frac{\alpha}{b_1}; \frac{\alpha}{b_1} + 1; \frac{(x - x_+)}{(x - x_-)} \right) \\
& + \left(\frac{\alpha}{\alpha + b_1} x_- e^{-(\alpha + b_1)t} \right) {}_2F_1 \left(1, \frac{\alpha + b_1}{b_1}; \frac{\alpha + b_1}{b_1} + 1; \frac{(x - x_+)}{(x - x_-)} \right) \\
& + (x_+) {}_2F_1 \left(1, \frac{\alpha}{b_1}; \frac{\alpha}{b_1} + 1; \frac{(x - x_+)}{(x - x_-)} e^{b_1 t} \right) \\
& \left. - \left(\frac{\alpha}{\alpha + b_1} x_- \right) {}_2F_1 \left(1, \frac{\alpha + b_1}{b_1}; \frac{\alpha + b_1}{b_1} + 1; \frac{(x - x_+)}{(x - x_-)} e^{b_1 t} \right) \right]^N
\end{aligned}$$

The hypergeometric factors are understood via principal-branch analytic continuation where the series argument leaves the unit disc (as occurs for subcritical parameters at large t); a full connection-formula treatment is omitted. Consistency checks include $G(1, 1, t) = 1$ and recovery of $G(x, y, 0) = y^N$. The absorption–birth–death closed form is algebraically heavy: intermediate integral coefficients involving $(s - x_+)/(s - x_-)$ must be tracked carefully through the substitution from characteristic coordinates, and the displayed expression should be treated as the formal outcome of that calculation pending independent numerical checks against the Kolmogorov system for small N .

One fact offers at least some consolation for the unwieldiness of this formula.

$${}_2F_1(1, \delta; \delta + 1; z) = \sum_{n=0}^{\infty} \frac{(1)_n (\delta)_n}{(\delta + 1)_n n!} z^n = \sum_{n=0}^{\infty} \frac{\delta}{\delta + n} z^n.$$

This holds because, along with $(1)_n = n!$, it can also be seen that

$$\frac{(\delta)_n}{(\delta + 1)_n} = \frac{\delta \cancel{(\delta + 1)} \cdots \cancel{(\delta + n - 2)} \cancel{(\delta + n - 1)}}{\cancel{(\delta + 1)} \cancel{(\delta + 2)} \cdots \cancel{(\delta + n - 1)} (\delta + n)} = \frac{\delta}{\delta + n}.$$

Path B result summary

For each of the three single-host models (absorption only; absorption–death; absorption–birth–death), the forward Kolmogorov equations compress to a first-order PDE for $G(x, y, t) = \mathbb{E}(x^{\mathbf{X}_t} y^{\mathbf{Y}_t})$.

The method of characteristics reduces that PDE to an ODE system along curves in (x, y) . Closed forms follow in the first two cases from elementary integrating factors; the third leads to explicit expressions involving Gauss hypergeometric functions, subject to the analytic-continuation caveat above. Coefficient extraction for multinomial-type generating functions is recorded in Appendix B. These calculations supply the template reused when later chapters add catastrophe transitions.

8 The constant $A(p)$ and the Koenigs obstruction

Path C returns to the discrete factor $A(p)$ introduced in Section 3.3. Path A already uses $A(p)$ as a well-defined limit, and already notes that, unlike its continuous counterpart A_{ct} , it is not given by a comparably elementary closed formula. The present section makes that statement precise enough for thesis work: we record usable representations, note the numerical and near-critical difficulties, and sketch why an elementary closed form in p is not expected, via the logistic map and its Koenigs linearisation. The heavier conjugacy arguments sit in Appendix C.

Earlier optimism in the project—that a tidy elementary expression for $A(p)$ might still turn up—is revised here. What replaces it is stronger and more useful: $A(p)$ is computable, has an infinite-product form, admits asymptotics as $p \rightarrow \frac{1}{2}^-$, and is tied to a classical object in one-dimensional dynamics for which elementary closed forms are known only outside the parameter range relevant to subcritical branching.

8.1 Recap: definition and conditional mean

For the death-or-division Galton–Watson process with division probability $p \in [0, \frac{1}{2})$,

$$S_{n+1} = 2p S_n - p S_n^2, \quad S_0 = 1,$$

and

$$A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n} \in (0, \tfrac{1}{2}], \quad (45)$$

so that

$$\lim_{n \rightarrow \infty} \mathbb{E}(\mathbf{Z}_n \mid \mathbf{Z}_n \neq 0) = \frac{1}{A(p)}.$$

Every analytic question about the limiting conditional mean is therefore a question about $A(p)$.

8.2 Numerical evaluation

For fixed p bounded away from $\frac{1}{2}$, iteration of the survival recurrence quickly drives S_n into the linear regime $S_{n+1} \approx 2p S_n$, so forming the ratio $S_n/(2p)^n$ for large n estimates $A(p)$ to arbitrary precision. The same computation yields the curve of $A(p)$ against p already compared with A_{ct} in Figure 2.

The method degrades as $p \rightarrow \frac{1}{2}^-$: $A(p) \rightarrow 0$, the conditional mean diverges, and the number of generations needed before S_n is small becomes impractically large. Near criticality one therefore needs an asymptotic surrogate (Section 8.4) rather than raw iteration alone.

8.3 Infinite product

A change of scale places the survival map in logistic form. Set $S_n = 2w_n$ and $r = 2p \in [0, 1)$. Then

$$w_{n+1} = r w_n (1 - w_n), \quad w_0 = \tfrac{1}{2},$$

which is the logistic map $f_r(w) = rw(1 - w)$ on the unit interval. Since

$$A(p) = 2 \lim_{n \rightarrow \infty} \frac{w_n}{r^n},$$

writing the limit as an infinite product of successive ratios $w_{k+1}/(rw_k)$ yields

$$A(p) = \frac{1}{2} \prod_{k=1}^{\infty} (1 - w_k) = \frac{1}{2} \prod_{k=1}^{\infty} \left(1 - \frac{S_k}{2}\right). \quad (46)$$

The product converges for $r \in (0, 1)$: the inequality $w_{n+1} < rw_n$ implies geometric decay of w_n , hence summability of $\sum w_k$ and convergence of $\prod(1 - w_k)$. Formula (46) is an exact representation rather than a closed elementary expression, since evaluating it still requires the orbit (w_k) generated by f_r . Its value is conceptual—it exhibits $A(p)$ as a functional of the logistic orbit—and computational, as an alternative running product alongside $S_n/(2p)^n$.

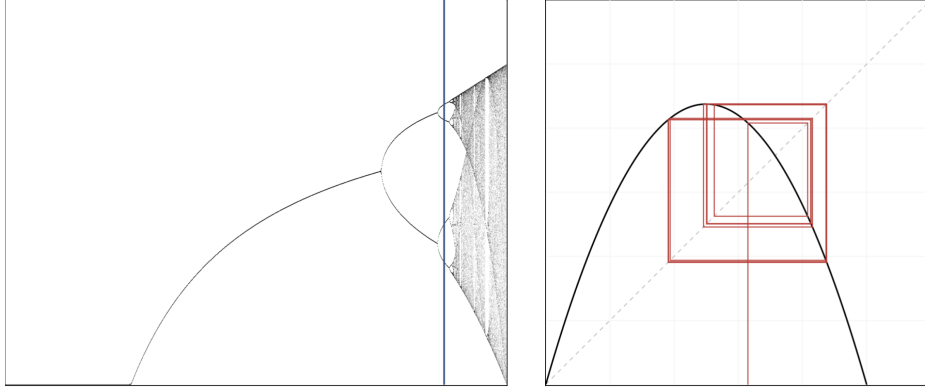


Figure 6: The logistic family $f_r(w) = rw(1 - w)$ for $r \in [0, 4]$ exhibits the familiar period-doubling cascade. Subcritical branching uses only the window $r = 2p \in [0, 1]$, far below the onset of chaos, but the same map appears in the product formula for $A(p)$.

8.4 Asymptotics near criticality

Let $\varepsilon = 1 - 2p \rightarrow 0^+$, so that $p = \frac{1}{2}(1 - \varepsilon)$ approaches criticality from below. The survival recurrence rearranges to

$$S_{n+1} - S_n = (2p - 1)S_n - pS_n^2 = -\varepsilon S_n - \frac{1}{2}S_n^2 + \frac{\varepsilon}{2}S_n^2.$$

Treating n as continuous for large n and dropping the higher-order term εS_n^2 produces the approximate ODE

$$\frac{dS}{dn} = -\varepsilon S - \frac{1}{2}S^2. \quad (47)$$

This Riccati equation is elementary. With $u = 1/S$ one obtains the linear ODE $u' - \varepsilon u = \frac{1}{2}$; matching $S_0 = 1$ and the late-time linearisation $S_n \sim A(p)(2p)^n$ then yields the leading near-critical behaviour

$$A(p) \sim 2\varepsilon = 2(1 - 2p) \quad \text{as } p \rightarrow \frac{1}{2}^-. \quad (48)$$

Consequently

$$\frac{1}{A(p)} \sim \frac{1}{2(1 - 2p)} \rightarrow \infty \quad \text{as } p \rightarrow \frac{1}{2}^-,$$

which agrees *exactly* at leading order with the continuous picture $1/A_{\text{ct}} = (1-p)/(1-2p) \sim 1/(2(1-2p))$ as $p \rightarrow \frac{1}{2}^-$. Formula (48) thus restores numerical access to $A(p)$ in precisely the window where direct iteration of S_n stalls.

8.5 Search attempts

Before the dynamical connection below was appreciated, a natural question was whether $A(p)$ might still be a simple combination of elementary functions of p . Direct attacks included continuum ODE approximations of the recurrence and, more experimentally, symbolic regression / genetic search for a closed expression fitted to numerical samples of $A(p)$ (cf. the genetic-algorithm experiments archived with the chapter materials). No satisfactory elementary candidate survived. Interactive illustrations of those searches are intended to be linked from the compiled PDF once hosted ([<URL: GA / symbolic regression demo>](#)). The negative outcome is consistent with—but weaker than—the Koenigs obstruction of the next subsections: failure to find a formula is not a proof, though it correctly predicted that the object is not elementary in p .

8.6 Logistic map and the Koenigs function

The substitution $S_n = 2w_n$, $r = 2p$ has already placed $A(p)$ inside the orbit of the logistic map f_r . Koenigs' classical linearisation theorem supplies a change of coordinates that straightens f_r near the attracting fixed point at the origin when $0 < |r| < 1$ [2]. There exists a unique local holomorphic map ϕ_r , with $\phi_r(0) = 0$ and $\phi'_r(0) = 1$, satisfying the Schröder equation

$$\phi_r(f_r(z)) = r \phi_r(z) \quad (49)$$

in a neighbourhood of 0. Equivalently,

$$\phi_r(z) = \lim_{n \rightarrow \infty} r^{-n} f_r^{\circ n}(z), \quad (50)$$

the limit existing uniformly on a neighbourhood of the origin.

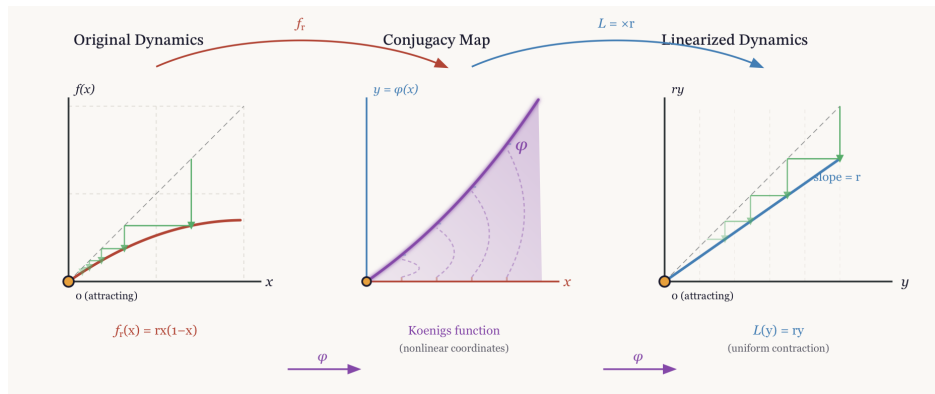


Figure 7: Schematic of Koenigs linearisation: the nonlinear logistic map f_r is conjugate, via ϕ_r , to multiplication by the multiplier r near the origin.

Comparing (50) with $A(p) = 2 \lim w_n / r^n$ and $w_n = f_r^{\circ n}(w_0)$, $w_0 = \frac{1}{2}$, one obtains the structural identity

$$A(p) = 2 \phi_r\left(\frac{1}{2}\right) \quad \text{with } r = 2p \in [0, 1), \quad (51)$$

interpreted in the sense of analytic continuation / the global real orbit on $(0, 1)$ under the attracting dynamics at 0 (the point $w_0 = \frac{1}{2}$ lies in the real basin for $r \in (0, 1)$; details and domain caveats are recorded in Appendix C). So $A(p)$ is not an arbitrary special function of p : it is twice the Koenigs function of the logistic map, evaluated at $\frac{1}{2}$, with multiplier $r = 2p$.

8.7 Why an elementary closed form is not expected

For each fixed multiplier $r \in (0, 1)$, the map f_r is a quadratic rational map with an attracting fixed point at 0 whose multiplier is not a root of unity. A theorem of Becker and Bergweiler [3] implies that the Koenigs linearisation of a rational map of degree at least two satisfies an algebraic differential equation if and only if the map is, up to analytic conjugacy, a monomial or Chebyshev map. For the quadratic family those exceptional cases correspond to logistic parameters $r = 2$ and $r = 4$ (Appendix C), both *outside* the interval $r \in (0, 1)$ relevant to subcritical branching.

For each fixed $p \in (0, \frac{1}{2})$, then, the linearisation of the associated logistic map is non-elementary as a function of its spatial argument, in the sense made precise in Appendix C (hypertranscendence of the conjugacy for multipliers in $(0, 1)$, with the classical elementary logistic cases $r = 2$ and $r = 4$ lying outside that interval). Extracting $A(p)$ from that conjugacy therefore cannot proceed by evaluating a single elementary expression for the conjugacy as a function of z .

Honest scope of the claim. Two logical gaps remain if one wants a theorem stronger than the working claim below. First, non-elementarity of a function $z \mapsto \phi(z)$ does not by itself forbid a closed form for a single value $\phi(\frac{1}{2})$ (compare Γ and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$); results on transcendence of *values* of local conjugacies would be needed to close that gap. Second, passing from fixed- r statements to a classification of the scalar map $p \mapsto A(p)$ is a further step. We do not claim either completion here. The working thesis claim, sufficient for applications and for revising the earlier “still attainable” hope, is:

No elementary closed-form expression for $A(p)$ on $p \in [0, \frac{1}{2})$ is known, and none is expected: $A(p)$ is tied to the linearisation of the logistic map at multiplier $r = 2p \in [0, 1)$, in a regime where that conjugacy is non-elementary for each fixed $r \in (0, 1)$, while the classical elementary logistic cases $r = 2, 4$ lie outside the subcritical branching range.

In that sense Path A’s punchline stands, Path C supplies mechanism and partial theorem, and $A(p)$ remains a perfectly usable numerical and asymptotic object.

Fractal aside. In the complex plane the same Koenigs conjugacies organise the local dynamics of quadratic maps whose global Julia sets are fractal. That geometric picture explains, informally, why one should not expect a simple elementary formula for the linearisation—but it is scenery, not a substitute for the Becker–Bergweiler dichotomy above.

8.8 Usable conclusions for the thesis

For the rest of the thesis one may treat $A(p)$ as:

1. well defined by (5) for each $p \in [0, \frac{1}{2})$;
2. computable by iteration or by the product (46), with asymptotics (48) near criticality;

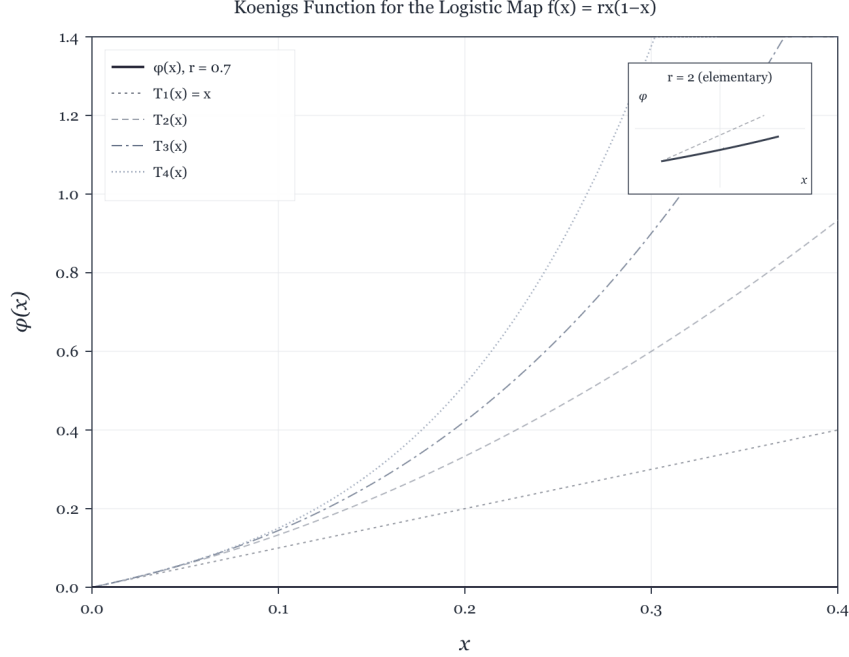


Figure 8: Numerical approximation of a Koenigs function for a logistic multiplier in $(0, 1)$, together with low-order Taylor polynomials near the origin. The conjugacy is smooth but not polynomial; elementary closed forms are available only for exceptional multipliers outside this range.

3. the reciprocal of the discrete quasi-stationary mean, in parallel with elementary A_{ct} in continuous time;
4. not replaceable, in the subcritical binary model, by a known elementary closed form in p .

Close of the chapter

Path A fixed the Markov, branching, and quasi-stationary language used throughout the thesis, including the continuous elementary counterpart of $A(p)$ and the sensitivity of survival to small founding cohorts. Path B developed the characteristic method for the continuous-time absorption generating functions used later in BDC analysis. Path C completed the discrete story of $A(p)$: representation, numerics, near-critical asymptotics, and a Koenigs-based account of why an elementary formula should not be expected. The appendices hold the heavier algebra. What remains for later chapters is application—not a missing elementary expression for $A(p)$.

A Appendix A-core: discrete second-moment derivation

This appendix records the second-moment calculation underlying (7) in Section 3.5. The notation matches the death-or-division Galton–Watson process with division probability $p \in (0, \frac{1}{2})$.

The offspring probability generating function is $\phi_1(z) = pz^2 + (1 - p)$, so the moment generating function of generation 1 is

$$M_1(t) = \phi_1(e^t) = pe^{2t} + (1 - p),$$

and moments satisfy $\mathbb{E}(\mathbf{Z}_n^r) = \frac{d^r}{dt^r} M_n(t)|_{t=0}$ when they exist. Writing $\phi_n = \phi_1^{\circ n}$ for the n -fold

iterate, the same holds at every generation with

$$M_n(t) = \phi_n(e^t).$$

Differentiating once along the iterate, by the chain rule applied through all n compositions, gives

$$M'_n(t) = (2p)^n M_{n-1}(t) M_{n-2}(t) \cdots M_1(t) e^{2t}. \quad (52)$$

Differentiating again, evaluating at $t = 0$, and using $M_k(0) = 1$ together with $M'_k(0) = (2p)^k$ produces

$$M''_n(0) = (2p)^n \left[\sum_{i=0}^{n-1} (2p)^i + 1 \right] = (2p)^n \left(\frac{1 - (2p)^n}{1 - 2p} + 1 \right) = (2p)^n \left(1 + \frac{1}{1 - 2p} \right) - (2p)^{2n} \left(\frac{1}{1 - 2p} \right),$$

which is (7). (The minus sign is required: at $n = 1$ the expression must return $\mathbb{E}(\mathbf{Z}_1^2) = 4p$.) The conditional limits (8)–(9) then follow from $S_n \sim A(p)(2p)^n$, exactly as in the main text.

B Appendix B-MoC: characteristic calculations

Heavy derivations supporting Section 7: integral identities, coefficient extraction, and special cases. Statement-level results remain in the main ladder.

Source appendix material

B.1 Deriving the integral result (43)

To demonstrate the result

$$\int \frac{e^{h\tau}}{p + qe^{k\tau}} d\tau = \frac{e^{h\tau}}{ph} {}_2F_1 \left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{q}{p} e^{k\tau} \right) + \text{const},$$

we will use the hypergeometric integral identity

$${}_2F_1(a, b; c; z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 u^{b-1} (1-u)^{c-b-1} (1-zu)^{-a} du \quad (53)$$

($\Gamma(x)$ being the standard gamma function).

Taking the integral,

$$I(\tau) = \int \frac{e^{h\tau}}{p + qe^{k\tau}} d\tau,$$

begin by applying the substitution $v = e^{k\tau}$, with $d\tau = k^{-1}v^{-1}dv$.

$$\begin{aligned} I(v(\tau)) &= \frac{1}{k} \int \frac{v^{\frac{h}{k}}}{p + qv} v^{-1} dv \\ &= \frac{1}{pk} \int \frac{v^{\frac{h}{k}-1}}{1 + \frac{q}{p}v} dv. \end{aligned}$$

Re-writing this as a definite integral from 0 to v using a dummy variable, s ,

$$I(v(\tau)) = \frac{1}{pk} \int_0^v \frac{s^{\frac{h}{k}-1}}{1 + \frac{q}{p}s} ds,$$

then changing variables again with $s = vu$ (as v is constant with regards to the definite integral, $ds = vdu$),

$$\begin{aligned} I(v(\tau)) &= \frac{1}{pk} \int_0^1 \frac{(vu)^{\frac{h}{k}-1}}{1 + \frac{q}{p}vu} v du \\ &= \frac{v^{\frac{h}{k}}}{pk} \int_0^1 \frac{u^{\frac{h}{k}-1}}{1 + \frac{q}{p}vu} du. \end{aligned} \tag{54}$$

This integrand lines up with that of the identity, (53);

$$\int_0^1 u^{\frac{h}{k}-1} (1-u)^0 \left(1 - \left(-\frac{q}{p}v\right)u\right)^{-1} du = \int_0^1 u^{b-1} (1-u)^{c-b-1} (1-zu)^{-a} du.$$

Notice, $b = \frac{h}{k}$, $c = \frac{h}{k} + 1$, $a = 1$, and $z = -\frac{q}{p}v$. Hence,

$$\begin{aligned} \int_0^1 \frac{u^{\frac{h}{k}-1}}{1 + \frac{q}{p}vu} du &= \frac{\Gamma\left(\frac{h}{k}\right) \Gamma(1)}{\Gamma\left(\frac{h}{k} + 1\right)} {}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{q}{p}v\right) \\ &= \frac{k}{h} {}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{q}{p}v\right), \end{aligned}$$

and, with (54),

$$I(v(\tau)) = \frac{v^{\frac{h}{k}}}{ph} {}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{q}{p}v\right).$$

Finally,

$$I(\tau) = \frac{e^{h\tau}}{ph} {}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{q}{p}e^{k\tau}\right) \quad (+\text{const}).$$

B.2 Extracting coefficients from the PGF – a special case

For a probability generating function defined not in terms of a series sum — for example,

$$G(t; x, y) = \left((1 - e^{-\alpha t})x + e^{-\alpha t}y \right)^n \tag{55}$$

— it is possible to make a conversion to its equivalent of the form

$$G(t; x, y) = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} p_{i,j}(t) x^i y^j, \tag{56}$$

thus extracting the time-dependent state probabilities, $p_{i,j}(t)$. All that is required is that the formula $G(t; x, y)$ can be partially differentiated continuously in x and y .

The reason is a bivariate generalisation of the Maclaurin expansion,

$$f(x, y) \approx Q(x, y) = \sum_{k=0}^{\infty} \left(\frac{1}{k!} \sum_{r=0}^{\infty} \binom{k}{r} \partial_x^r \partial_y^{k-r} f \Big|_{(0,0)} x^r y^{k-r} \right).$$

The coefficients of (56) can hence be generated via

$$p_{i,j}(t) = (i!j!)^{-1} \partial_x^i \partial_y^j f|_{(0,0)}. \quad (57)$$

A special case

For a generating function of multinomial type,

$$G(t; x, y) = (c(t) + f(t)x + g(t)y)^n,$$

the multinomial theorem gives, whenever $i + j \leq n$,

$$p_{i,j}(t) = \binom{n}{i, j, n-i-j} f(t)^i g(t)^j c(t)^{n-i-j}, \quad (58)$$

and $p_{i,j}(t) = 0$ otherwise. (In particular the incorrect product rule $\partial_x^a \partial_y^b G = (nf)^a (ng)^b G$ must not be used.)

Applied to PGFs from absorption only and absorption-death

For the absorption-only example (55), one has $c \equiv 0$, $f = 1 - e^{-\alpha t}$, $g = e^{-\alpha t}$, supported on $i + j = n$:

$$p_{i,j}(t) = \binom{n}{i} (1 - e^{-\alpha t})^i (e^{-\alpha t})^j \quad (i + j = n).$$

The absorption–death generating function is again of multinomial type, with coefficients read from the closed form of G ; its state probabilities are the corresponding multinomial weights, not the formula with factors $n^{i+j}/(i!j!)$ that appeared in an earlier draft.

C Appendix C-Koenigs: conjugacy details and exceptional cases

This appendix supports Section 8. It records the affine conjugacy between the logistic and quadratic families, the Becker–Bergweiler dichotomy, the elementary solutions at $r = 2$ and $r = 4$, and domain remarks for the identity $A(p) = 2\phi_{2p}(\frac{1}{2})$.

C.1 Affine conjugacy to $z^2 + c$

Lemma C.1. *The logistic map $f_r(x) = rx(1 - x)$ is affinely conjugate to $P_c(z) = z^2 + c$ with*

$$c(r) = \frac{2r - r^2}{4}. \quad (59)$$

Proof sketch. Seek $h(x) = ax + b$ with $h \circ f_r \circ h^{-1} = P_c$. The standard choice $x = -\frac{z}{r} + \frac{1}{2}$ (for $r \neq 0$) conjugates f_r to a monic quadratic $z^2 + c$; matching coefficients yields (59). $\square$

Under $r \in (0, 1)$ one has $c(r) \in (0, \frac{1}{4})$. The exceptional integrable quadratics below correspond to $c = 0$ ($r = 2$) and $c = -2$ ($r = 4$), both outside $(0, \frac{1}{4})$.

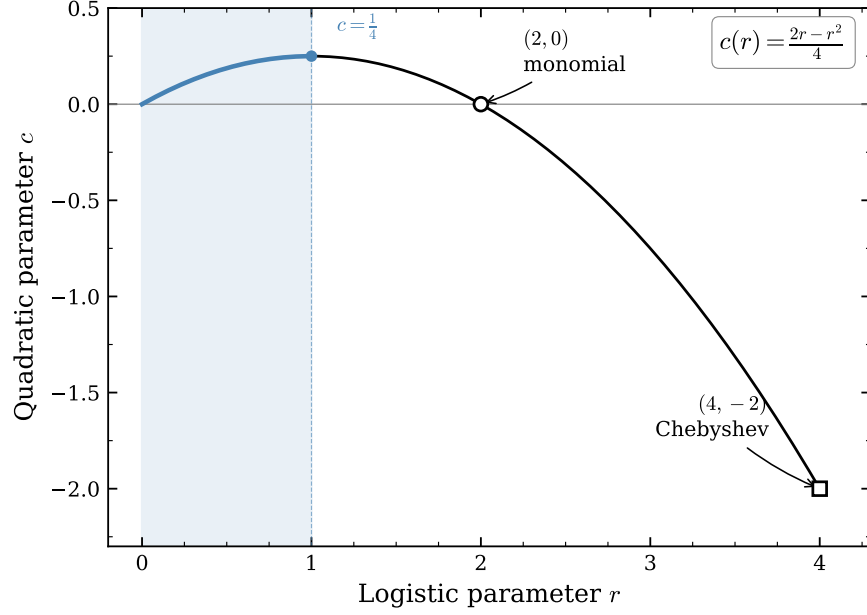


Figure 9: Parameter correspondence $c(r) = (2r - r^2)/4$. The segment $r \in (0, 1)$ maps to $c \in (0, \frac{1}{4})$, disjoint from the integrable values $c \in \{0, -2\}$.

C.2 Non-elementarity for multipliers in $(0, 1)$

For $r \in (0, 1)$ the origin is an attracting fixed point of f_r with multiplier $\lambda = r \in (0, 1)$, and classical linearisation (Koenigs) supplies a holomorphic conjugacy ϕ_r near 0 satisfying (49). Results of Becker and Bergweiler on hypertranscendence of conjugacies in complex dynamics [3] imply that, for rational maps of degree at least two, such linearisations are differentially algebraic only in highly exceptional normal forms (monomial / Chebyshev type). Under the conjugacy of Lemma C.1, the interval $r \in (0, 1)$ maps to $c \in (0, \frac{1}{4})$, which avoids those exceptional quadratic parameters. Hence for each fixed $r \in (0, 1)$ the conjugacy ϕ_r is non-elementary as a function of z .

Remark C.2 (Solvable logistic parameters outside the branching range). The logistic parameters $r = 2$ and $r = 4$ are classically solvable (logarithmic and inverse-trigonometric linearisations, up to normalisation), corresponding under (59) to $c = 0$ and $c = -2$. At those parameters the multiplier at 0 is *repelling* ($|r| > 1$), so the linearisation is of Poincaré rather than attracting-Koenigs type. They illustrate that elementary formulae exist only for exceptional multipliers *outside* $r \in (0, 1)$; they are not counter-examples inside the subcritical branching window $r = 2p \in [0, 1)$.

C.3 Elementary displays at $r = 2$ and $r = 4$ (outside Path C range)

Case $r = 2$ ($c = 0$). A standard normalisation yields a logarithmic linearisation equivalent to $-\frac{1}{2} \log(1 - 2z)$ on a suitable domain.

Case $r = 4$ ($c = -2$). A standard normalisation is proportional to $(\arcsin \sqrt{z})^2$.

Neither formula supplies an elementary closed form for general $r \in (0, 1)$.

C.4 On $A(p) = 2\phi_{2p}(\frac{1}{2})$

With $w_{n+1} = f_r(w_n)$, $r = 2p$, $w_0 = \frac{1}{2}$, the Koenigs limit (50) gives

$$\phi_r(\frac{1}{2}) = \lim_{n \rightarrow \infty} r^{-n} f_r^{\circ n}(\frac{1}{2}) = \lim_{n \rightarrow \infty} \frac{w_n}{r^n},$$

whenever the orbit stays in the domain where the limit representation holds. For $r \in (0, 1)$ the origin is attracting on the real line and $w_0 = \frac{1}{2}$ iterates towards 0 under f_r , so the real orbit is the natural candidate for the limit. Then

$$A(p) = 2 \lim_{n \rightarrow \infty} \frac{w_n}{r^n} = 2\phi_r(\frac{1}{2})$$

as stated in (51). The endpoint $p = 0$ ($r = 0$) is elementary and degenerate (immediate extinction after the first generation in the binary model with $p = 0$); it may be checked directly that $A(0) = \frac{1}{2}$ matches $\frac{1}{2} \prod_{k \geq 1} (1 - w_k)$ with $w_k = 0$ for $k \geq 1$ after the first step, consistently with $A(p) \leq \frac{1}{2}$.

C.5 Parameter-dependent versus fixed- r statements

Fixed- r non-elementarity constrains ϕ_r as a function of z , whereas the thesis uses the scalar $A(p) = 2\phi_{2p}(\frac{1}{2})$. Even for a single r , non-elementarity of $z \mapsto \phi_r(z)$ does not by itself forbid a closed form for one value $\phi_r(\frac{1}{2})$; and elevating the statement to a classification of $p \mapsto A(p)$ is a further step. The main text therefore adopts the careful claim quoted in Section 8.7.

References

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